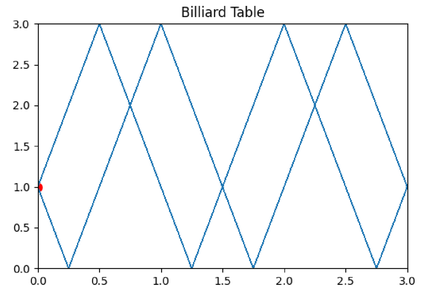
Billiards and Periodic Orbits

Tuong Thuc Le Dee He Kasandra Bahena
University of Illinois at Chicago

MSCS Undergraduate Research Laboratory

Summary

What happens when a ball bounces around a frictionless, pocketless billiard table? A classic problem in physical and chemical sciences is the trajectories and collisions of particles inside a contained environment [1]. The billiard model studies this with frictionless, pocket-less tables of any shape—here we focus on squares.



Motivation

We examine periodic orbits that return to their starting point, analyzing their periods and slopes. Using cutting sequences and computer programs, we draw conclusions about the topological entropy of these systems within ergodic theory.

Angle of Incidence & Reflection

Diving into the most important mechanics to understanding this research: the angle at which it approaches a side (the angle of incidence) is equal to the angle at which it leaves (the angle of reflection):

$$\theta_i = \theta_r$$

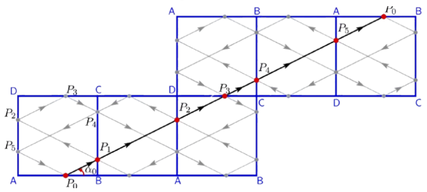
Both angles are measured relative to the edge at the point of contact, where **n** is perpendicular to the boundary (**n** $\perp$ edge).

Unfolding

A commonly used technique in this study called *unfolding* [2] transforms a billiard trajectory into a straight line on a singular plane, instead of the trajectory being reflected. Understanding *unfolding* is as follows:

$$y = \tan(a_0)(x - P_0)$$

Where P_0 is the initial point on a billiard table's side (eg. $P_0 \in [0,3)$) and a_0 is the initial angle of the ball's launch.

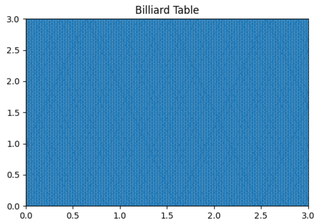


Periodic & Aperiodic Orbits

Having mentioned of periods, in billiard dynamics, the trajectory of a ball can be classified as either **periodic** or **aperiodic** [3]:

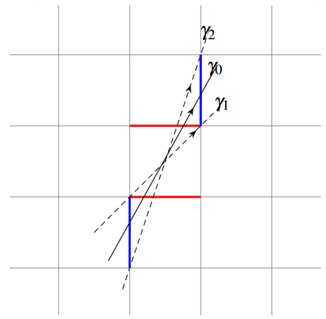
- **Periodic orbit:** There exists $n \in \mathbb{N}$, for all $n > 0$, such that after n reflections, the ball returns to its initial position and direction.
- **Aperiodic orbit:** Conversely, for all $n > 0$, the ball never exactly repeats its path or state.

Understanding whether an orbit is periodic or aperiodic is fundamental to analyzing the complexity and behavior of billiard systems. Below is a generated image showing an instance of an aperiodic orbit, where the trajectory left behind from the ball covers the majority of the table.



Cutting Sequences

One way to study the complexity of the behavior of a path is by analyzing the patterns of its "cutting sequence". To understand what a cutting sequence is, let us look at an example:



Here, a small portion of an arbitrary path γ_0 is embedded in a grid. Whenever it intersects with a horizontal/vertical line, that particular segment is colored red/blue, and we record those instances with A/B, respectively.

Therefore, these four instances shown in the figure above can be represented as *BAAB*, and the **cutting sequence of** γ_0 is simply the collection of all such instances. Further, we can write $BAAB \in \mathscr{L}$ to indicate that the "word" *BAAB* is in the cutting sequence.

Conversely, if $BAAB \in \mathscr{L}$, it can be observed that the path can be as flat as γ_1, and as steep as γ_2, i.e., if we let d be the slope of the path, then $1 < d < 3$. (Note: the inequities are strict here because we want to avoid the trickiness of encountering corners — in this particular case, it can be difficult to determine how to record them in a cutting sequence.) In fact: if $BA^nB \in \mathscr{L}$, then $n-1 < d < n+1$. And it follows easily that: if $BA^nB \in \mathscr{L}$ and $BA^mB \in \mathscr{L}$, then $m \in \{n-1, n, n+1\}$.

A Periodic Orbit of Three Does Not Exist

Assume for contradiction that there is a triangular path of incidence within a square billiard table.

Consider one side of the triangle representing an incidence ray that strikes a flat wall. By the law of reflection, the angle of incidence equals the angle of reflection ($\alpha_1 = \alpha_2$). Additionally, let the corresponding exterior angles be A_1 and A_2. Since the interior angles are equal, the exterior ones are also equal ($A_1 = A_2$). All four angles lie along a straight line:

$$2\alpha + 2A = 180^\circ$$

Divide both sides:

$$\alpha + A = 90^\circ$$

Now consider the second incidence path formed by the reflection of the first. Let the new incidence and reflection angles be β_1 and β_2 and the corresponding exterior angles be C_1 and C_2. These four angles also lie on a straight line:

$$\beta_1 + \beta_2 + C_1 + C_2 = 180^\circ$$

$$\beta + C = 90^\circ$$

From the small triangle formed by the exterior angles A and C and the interior angle γ:

$$A + C + \gamma = 180^\circ$$

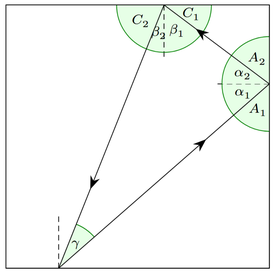
However, applying the reflection law to the straight edges yields two key equalities:

$$\alpha + A = 90^\circ \quad \text{and} \quad \beta + C = 90^\circ$$

Through substitution and geometric constraints ($A = \beta$ and $C = \alpha$), the angle sum must also satisfy $A + C = 90^\circ$. Substituting this required condition into the triangle's angle sum ($A + C + \gamma = 180^\circ$):

$$90^\circ + \gamma = 180^\circ \implies \gamma = 90^\circ$$

This forces a final substitution into the overall sum of the path's reflections, leading directly to the contradiction that $\gamma = 0^\circ$ for the path to be a triangle. If $\gamma = 0$, the path collapses into a straight line, which violates the initial assumption of a triangular path.



$\#\mathscr{L}$ with length n's pattern is $n+1$, $\forall n \in \mathbb{N}$

We conjecture that for cutting sequences generated by natural number slopes, the number of distinct words of length n is exactly $n+1$. This can be proven in the case $n = 2$. For general n, we provide computational evidence below.

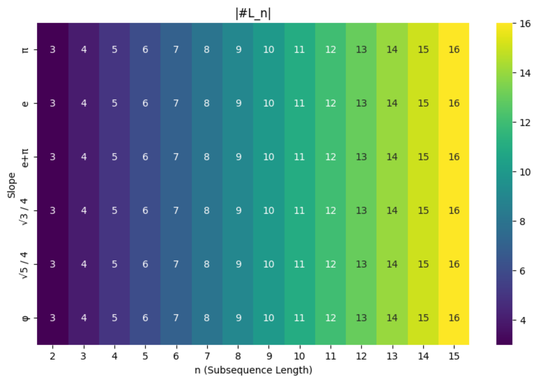
$$\#\mathscr{L}_n = n+1, \quad \forall n \in \mathbb{N}, \forall s \in \mathbb{N}$$

Computational Evidence for $\#\mathscr{L}_n = n+1$

We can verify the conjecture computationally for various natural number slopes and values of n. The pattern consistently shows that the complexity function grows linearly with slope 1.

For $n = 1$, with A being the vertical bounce and B being horizontal, we have exactly 2 distinct outcomes of length 1: $\{A, B\}$, so $\#\mathscr{L}_1 = 2 = 1+1$.

The Python implementation below demonstrates this pattern for multiple values of n and confirms the conjecture holds experimentally:



Topological Entropy

From what we know about $\mathscr{L}$ and $\mathscr{L}_n$ so far, we can also define and compute the **topological entropy** of a cutting sequence X (always equates to zero):

$$\begin{aligned} h_{top}(X) &:= \lim_{n\to\infty} \frac{1}{n} \log |\mathscr{L}_n| \\ &= \lim_{n\to\infty} \frac{\log(n+1)}{n} \\ &= \lim_{n\to\infty} \frac{1}{n+1} \quad \text{(L'Hôpital's Rule)} \\ &= 0 \end{aligned}$$

Project supervisor & Acknowledgments

A big thank you to our project mentor and supervisor Professor Benjamin Call for all their help and all they taught us. Additionally, we'd also like to greatly thank Professor Daniel Groves for organizing MURL Fall '25, and Diana Davis, whose work greatly influenced us.

References

[1] N. Chernov, R. Markarian. *Chaotic Billiards* (2006), ix.

[2] D. Davis, *The Geometry of Billiards*, (2020).

[3] Hongjia H. Chen and Hinke M. Osinga, *Periodic orbits for square and rectangular billiards*, (2024), i.